

How predictive is diffusion MRI microstructure?

A reproducible benchmark across datasets, features and models

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Abstract. Diffusion MRI is widely used to explore brain microstructure, and numerous machine learning methods have been proposed to extract predictive information. Yet progress remains difficult to assess: studies rely on single datasets, heterogeneous pipelines, limited evaluation, and often lack accessible code, preventing meaningful comparison and reproducibility. We introduce a modular and transparent benchmarking framework for machine learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models. It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods. This shows that standardized benchmarking is essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance. We introduce a modular and transparent benchmarking framework for machine learning in dMRI microstructure. The benchmark unifies multiple representative datasets within a standardized end-to-end pipeline and enables systematic comparison across tissue types, feature representations and models. It reveals differences between white and gray matter analyses, quantify the impact of diffusion microstructure feature choices, and show that fast linear baselines can rival more complex deep learning methods, demonstrating that standardized benchmarking is now essential to move the field beyond proxy tasks and toward robust methods with real clinical relevance.

Keywords: Benchmark · Brain · diffusion MRI. **[TODO: Add keywords on ML]**

1 Introduction

Diffusion magnetic resonance imaging (dMRI) enables the in vivo representation of tissue microstructure and provides biologically meaningful biomarkers sensitive to axonal organization, myelination, and tissue complexity [19]. Leveraging these microstructural descriptors for predictive modeling has become increasingly common, with applications spanning demographic estimation [4, 15],

clinical diagnosis [11, 29], and cognitive characterization [17]. The combination of diffusion-derived features and machine learning is therefore often viewed as a powerful framework for data-driven neurobiological inference.

Comparing predictive models across diffusion MRI studies remains challenging. Preprocessing pipelines differ substantially, microstructural information is represented in multiple ways (e.g., voxel-wise maps, ROI summaries, or learned embeddings), and evaluation protocols vary in data splits, cross-validation schemes, and hyperparameter selection. As a result, reported performance differences may reflect methodological inconsistencies rather than genuine modeling improvements. Limited code and configuration sharing further restrict reproducibility. Together, these factors make it difficult to determine which approaches generalize reliably and provide meaningful insights.

While many report promising predictive power, reported performance varies widely across studies, tasks, and datasets. Direct comparison between studies is often hard to achieve. Improvements are frequently attributed to model complexity or architectural innovation, yet it remains unclear to what extent these gains arise from representational choices, preprocessing variability, or evaluation design. As the number of dMRI-based prediction studies grows, the absence of a standardized and reproducible evaluation framework has become a critical bottleneck. Without controlled evaluation across datasets, preprocessing pipelines, and model families, it is difficult to assess methodological progress and understand the limitations of current approaches.

To address this problem, we introduce an open-source diffusion MRI benchmark that enables controlled comparison across datasets, tissue types, microstructural representations, and model families. The benchmark explicitly quantifies performance variability across folds and configurations, isolates representation and tissue-specific effects, and evaluates the value of diffusion-derived features beyond non-diffusion baselines. We believe this provides the foundation for isolating microstructural signal from configuration-driven variability and for establishing comparable evaluation standards in diffusion MRI prediction.

2 Benchmark

[TODO: Find a better section title] In this work, we introduce a standardized benchmark that unifies preprocessing and feature extraction, evaluates both classical and deep learning models on regression and classification tasks over a rigorous evaluation protocol on multiple public datasets. The overall structure is illustrated in Figure 1.

2.1 Datasets

To ensure reproducibility, we evaluate our benchmark on three publicly available datasets covering both healthy and clinical populations. **HCP**: The Human Connectome Project (HCP) Young Adult dataset [26] targets gender and age. It includes 1,036 healthy subjects after filtering (474 males, 562 females;

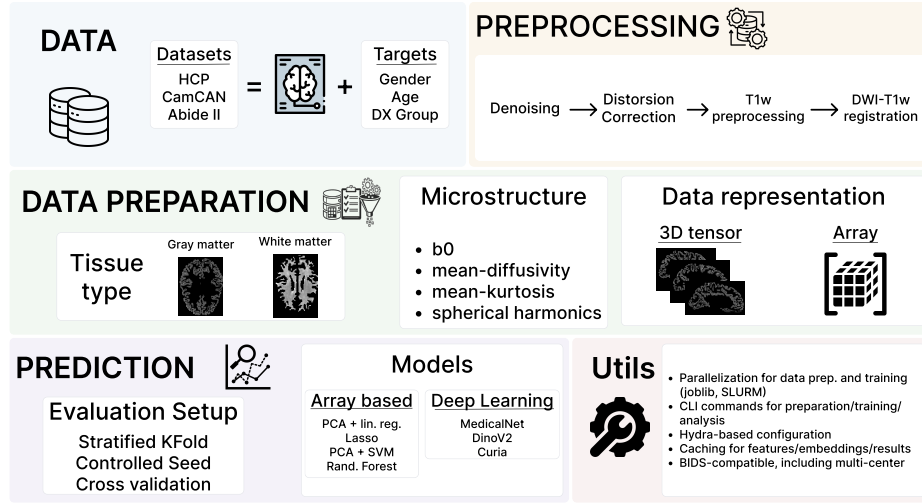


Fig. 1. Overview of the proposed diffusion MRI benchmark pipeline, illustrating the end-to-end process from preprocessing to microstructural feature extraction, predictive modeling, and evaluation.

ages 22-37). HCP data were acquired on a customized Siemens 3T Connectome scanner using a multi-shell Spin-Echo EPI sequence (TR/TE = 5520/89.5 ms, 1.25 mm isotropic, multiband factor 3; b -values = 1000, 2000, 3000 s/mm², 90 directions/shell). **Cam-CAN**: The Cambridge Centre for Ageing and Neuroscience (Cam-CAN) dataset [22] targets gender and age. It includes 629 healthy subjects (316 males, 313 females; ages 18-88), scanned on a Siemens 3T TIM Trio using a 2D twice-refocused SE EPI sequence (TR/TE = 9100/104 ms, 2.0 mm isotropic; b -values = 1000, 2000 s/mm², 30 directions each). **ABIDE II**: The Autism Brain Imaging Data Exchange II (ABIDE II) dataset [10] targets Autism Spectrum Disorder (ASD) diagnosis. It includes 218 subjects (186 males, 32 females; ages 5-64), including 119 with ASD and 99 controls. As a multi-site dataset, ABIDE II data were acquired on various 3T scanners with highly heterogeneous, site-specific single-shell diffusion protocols. Across the included sites, spatial resolutions range from 0.94 to 3.0 mm, TR/TE from 5200-20244/78-101 ms, b -values include 1000, 1500, and 2500 s/mm², and diffusion directions vary between 32 and 64. This variability provides a realistic scenario for evaluating model robustness to scanner heterogeneity.

2.2 Data Preparation

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Preprocessing Except for HCP, which is already preprocessed, DWI data are preprocessed using the `diffusion-preprocessing` package¹ which containerizes various tools such as `dipy` [12], `sMRIPrep` [?], `FSL` [24], `ANTs` [2], `FreeSurfer` [8] and `AFNI` [7]. The images are first denoised using the Marchenko-Pastur PCA (MPPCA) method [28, 6] implemented in `dipy`. This denoising procedure identifies the principal components corresponding to noise and then removes them from the DWI data [28]. After MPPCA denoising, Gibbs ringing artifacts [27] are also removed using `dipy`. Next, to correct the distortions due to inhomogeneities of the magnetic field, `FSL`’s eddy correction [1] is used. Finally, this denoised and distortion-corrected DWI data are registered to the anatomical T1w image, which was preprocessed using `sMRIPrep`, using `FreeSurfer`’s `bbregister` [13] with 12 degrees of freedom and boundary-based registration cost function.

Microstructural representation To characterize the biological properties of brain tissue, we compute a diverse set of microstructural measures using established diffusion models. These measures capture complementary aspects of tissue architecture, ranging from overall diffusion magnitude to more complex ones:

- **Baseline ($b = 0$):** The averaged non-diffusion-weighted (T2w) signal is included as a reference to disentangle diffusion-driven predictive effects from underlying structural T2 contrast.
- **Mean Diffusivity (MD):** MD is computed via Diffusion Tensor Imaging (DTI) [3], and reflects the overall magnitude of water diffusion.
- **Mean Kurtosis (MK):** Mk is derived from Diffusion Kurtosis Imaging (DKI) [16], quantifies deviations from Gaussian diffusion and acts as a proxy for microstructural heterogeneity and tissue complexity.
- **Spherical Harmonics (SH) Power:** SH is a model-free representation obtained by computing the L2 norm of the SH coefficients [25] (up to order 6) fitted directly to the diffusion signal (up to order 6). This representation especially captures signal anisotropy without assuming a specific compartmental model.

To mitigate free-water effects and scanner-dependent intensity scaling [21], we apply ventricular normalization: for each subject and each microstructural representation, voxel-wise diffusion measures are divided by the mean value extracted from cerebrospinal fluid within the ventricles. Finally, extreme values are clipped at the 99th percentile to ensure numerical stability during model training.

Spatial normalization To enable anatomically consistent comparisons across subjects and datasets while respecting tissue-specific geometry, we adopt distinct spatial normalization strategies for gray and white matter.

¹ <https://github.com/man-shu/diffusion-preprocessing>

Gray Matter. Gray matter microstructure is represented on the subject-specific cortical surface reconstructed from T1w images using FreeSurfer. Cortical meshes (white and pial surfaces) are generated using the Desikan–Killiany atlas. Diffusion scalar maps are computed in the native diffusion space and rigidly aligned to the anatomical image. Voxelwise diffusion measures are then projected onto the individual cortical surface using ribbon-constrained sampling, preserving vertex-wise resolution (64k vertices per hemisphere) [13]. This surface-based strategy respects cortical geometry, reduces partial-volume effect, maintains high spatial specificity and enables consistent spatial normalization across subjects.

White Matter White matter microstructure is represented in a standardized tract space using Tract-Based Spatial Statistics (TBSS) [23]. Diffusion scalar maps are nonlinearly registered to standard space and projected onto a group-wise white matter skeleton, which captures the core of major fiber bundles and minimizes residual misalignment and partial-volume effects. Microstructural values are then summarized using the JHU White-Matter Tractography Atlas [18], extracting mean values from 48 major tracts. This atlas-constrained, skeleton-based normalization enforces anatomical correspondence across subjects and reduces variability associated with subject-specific tractography. While this approach sacrifices voxel-level granularity compared to the cortical surface representation, it provides a compact, reproducible, and anatomically interpretable feature set that is particularly robust in multi-site settings.

2.3 Prediction models

Our benchmark includes a wide range of predictive models, including linear and non-linear models, classical machine learning pipelines, and pre-trained foundation models within a unified evaluation framework. Specifically, linear models (e.g., logistic regression, Lasso), Random Forests [TODO: Add the quote for random forests (rf paper from 2000 something)], and Support Vector Machines (SVM) [14]. Given the high dimensionality of the microstructural representations, we additionally assess these models combined with Principal Component Analysis (PCA) for dimensionality reduction.

We further benchmark several vision-based deep architecture. DINOv2 [20] and Curia [9], are used as frozen feature extractors with prediction heads trained on top. In contrast, the pretrained 3D convolutional architecture MedicalNet [5] is fine-tuned on our diffusion datasets to adapt to domain-specific microstructural patterns.

2.4 Benchmarks tasks and evaluation protocol

The prediction tasks covers both classification and regression and were selected based on clinical relevance, data availability, and existing literature.

Classification We considered binary classification tasks, including Autism Spectrum Disorder (ASD) diagnosis (ASD vs. Typically Developing) using ABIDE II, and gender prediction in the HCP and Cam-CAN cohorts. Performance is reported using balanced accuracy, which accounts for class imbalance.

Regression For regression, we evaluate brain age prediction using the healthy Cam-CAN cohort. Performance is quantified using the coefficient of determination (R^2) and mean absolute error (MAE).

2.5 Evaluation protocols

To ensure rigorous and reproducible model assessment, we employ a stratified 5-fold cross-validation strategy for all tasks. Stratification maintains balanced target distributions across folds in classification settings and in regression tasks, gender is used. Data splits are fixed through seed initialization to guarantee full reproducibility. Strict separation between training and test data is enforced at every stage.

All preprocessing steps involving data-driven operations (e.g., feature scaling, PCA, and hyperparameter tuning) are fitted exclusively on the training portion of each fold and subsequently applied to the held-out test set.

3 Results

We evaluated the predictive power of diffusion MRI microstructural representations across datasets, tasks, tissue compartments, and model families. Rather than focusing only on peak performance, we analyze global patterns across configurations, allowing us to assess performance stability, tissue-specific contributions, and the added value of diffusion-derived features beyond non-diffusion baselines.

Figure 2 highlights two central findings. First, when evaluated on a common normalized scale, performance differences between model families are consistently modest. Regularized linear models remain competitive with deep learning architectures, independently of the task and data preparation strategy. Thus, the current data representation may not be sufficient to leverage the full potential of recent neural architectures. This likely reflects dataset size constraints; image-based models (Curia, DinoV2) were used as frozen feature extractors without fine-tuning, which typically demands larger cohorts to yield competitive performance.

Second, variability across preprocessing and microstructural parameterizations is substantial. In several tasks, the dispersion induced by feature and tissue configurations is larger than the differences observed between model families. This sensitivity indicates that predictive performance in diffusion MRI cannot be attributed to model choice alone. Instead, preprocessing strategy happens to be a major and often underreported source of variance. Conclusions drawn from a single representation or configuration may be unstable, therefore, underscoring the need for robustness analyses across preprocessing settings.

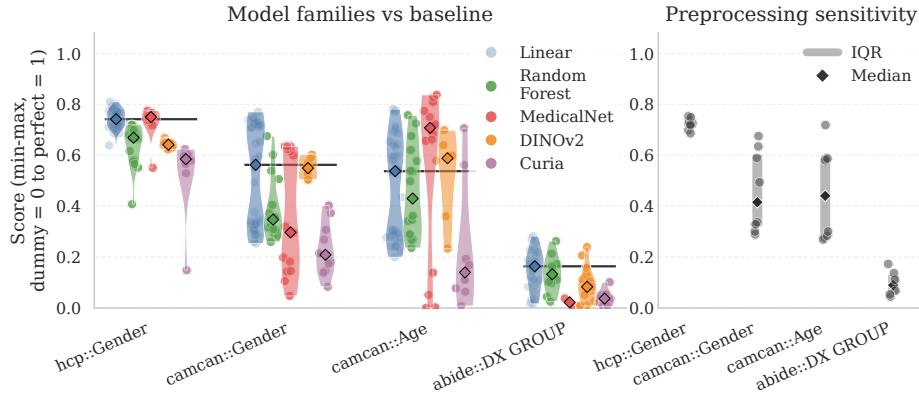


Fig. 2. Combined analysis of model-family performance and preprocessing sensitivity across tasks using a normalized score (0 = dummy, 1 = perfect) for cross-metric comparison. Left: Cross-validation results grouped by model family (points = models, violins = distributions, diamonds = medians; lines = linear baseline). Right: Sensitivity to preprocessing across feature $\times$ tissue configurations (dots = strategies; bars = IQR and range).

3.1 Impact of tissue type

We next quantify the impact of tissue type by comparing the performance of models with white and gray matter under identical dataset–target–task–feature configurations. For each fold, performance is normalized relative to a dummy baseline and the difference $\Delta = \text{white} - \text{gray}$ is computed. To disentangle tissue influence from dataset and representation biases, we residualize the effects in two complementary ways: (i) removing feature-specific baselines to isolate dataset-dependent tissue effects, and (ii) removing dataset–task baselines to isolate feature-dependent tissue effects. Statistical inference is performed on configuration-level mean effects to account for fold dependence, testing whether the tissue difference distribution deviates from zero.

As shown in Figure 3, tissue effects are strongly dataset-dependent. In HCP, we observe a systematic predominance of white matter over gray matter, with positive effect distributions indicating consistent white-matter superiority across configurations. In contrast, CamCAN exhibits the opposite pattern, with gray matter showing a stable advantage. For ABIDE, the distribution of differences is centered around zero, suggesting no clear predominance of either compartment.

When stratified by microstructural representation, most features do not demonstrate a consistent tissue preference. Overall, these results indicate that tissue effects are primarily dataset-dependent rather than driven by a specific microstructural representation.

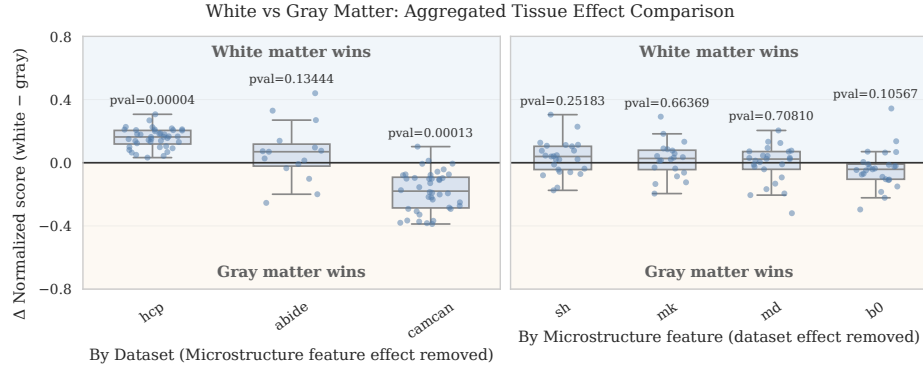


Fig. 3. Fold-level paired comparison of white vs. gray matter using normalized scores (0 = dummy, 1 = perfect). Each point shows the matched fold difference (Δ = white – gray) under identical settings. Left: tissue effects by dataset $\times$ target (microstructure baseline removed); Right: tissue effects by microstructure feature (dataset-task baseline removed). Boxplots summarize fold differences; p-values test mean effects against zero.

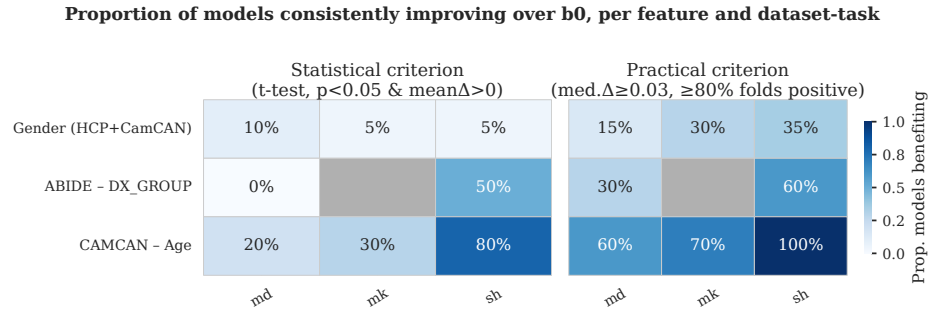


Fig. 4. Proportion of models improving over the non-diffusion baseline (b0), stratified by task and feature. Improvement is based on fold-level normalized differences (Δ = feature – b0). Left: statistical criterion (t-test against zero, $p < 0.05$ and $\text{mean}\Delta > 0$); Right: robustness criterion (median $\Delta \geq 0.03$ and $\geq 80\%$ positive folds). Cells report the fraction of models meeting each criterion per target (rows) and feature (columns).

3.2 Impact of diffusion features

To assess whether diffusion-derived microstructural features provide predictive value beyond non-diffusion signal, we quantified model-wise improvements relative to b_0 . The results reveal that diffusion information does not uniformly enhance performance across tasks or datasets. For proxy tasks such as gender classification in HCP and CamCAN, diffusion features do not confer a systematic advantage over b_0 , indicating that non-diffusion signal (likely reflecting global anatomical properties such as brain size) already captures most of the discriminative information. In contrast, tasks more directly linked to microstructural variation show measurable gains from diffusion metrics. Age prediction in CamCAN and diagnostic classification (DX_GROUP) in ABIDE exhibit consistent improvements for several diffusion representations, particularly under linear models operating on array-based features (e.g., spherical harmonics).

Notably, deep learning models tend to exhibit positive gains in certain gender classification settings, likely reflecting their sensitivity to global morphological cues embedded in image-based representations rather than diffusion-specific contrast. Overall, these findings indicate that the added value of diffusion MRI depends strongly on the biological relevance of the task, the dataset characteristics, and the modeling framework, rather than representing a universal performance boost.

4 Conclusion

We introduced a diffusion MRI benchmark for demographics prediction in individuals across three datasets: Human Connectome Project, Cambridge Centre for Ageing and Neuroscience, and Autism Brain Imaging Data Exchange. Within a unified experimental framework, we systematically compared multiple microstructural representations (b_0 , MD, SH, MK) and model families spanning from linear to deep learning models. By controlling preprocessing, model selection, and evaluation procedures, the benchmark enables comparison across configurations and reduces methodological heterogeneity simplifying more transparent and reproducible comparisons in diffusion MRI prediction studies.

Our results suggest that predictive performance in diffusion MRI is highly configuration-dependent. In several settings, regularized linear models performed similarly to more complex approaches. More importantly, variability across folds, tissue types, and microstructural representations was often comparable to, or larger than, mean performance differences. Tissue type preferences were task-specific, and most microstructural features do not exhibit universal superiority. Diffusion-derived metrics provide measurable gains only in tasks closely aligned with microstructural variation, whereas for proxy tasks, non-diffusion signal can be equally informative.

This benchmark should not be interpreted as a definitive comparison of models or features, but rather as a controlled reference point. The included datasets, although widely used, remain moderate in size for modern machine learning,

especially for deep learning approaches that typically require larger cohorts or fine-tuning. In addition, the current analysis focuses on within-dataset evaluation. The ability of models and representations to generalize across cohorts, scanners, and acquisition protocols remains an open question. This aspect was intentionally left outside the scope of the present work in order to maintain a clear focus on controlled within-dataset evaluation.

Several directions for future work follow naturally from these limitations. First, expanding the benchmark to include additional and larger datasets would improve statistical power and allow stronger conclusions, particularly for deep learning methods. Second, incorporating more clinically focused datasets with well-defined diagnostic outcomes could help clarify the clinical relevance of diffusion-derived features. Third, harmonization strategies across scanners and sites should be systematically evaluated, as variability in acquisition and preprocessing may substantially influence predictive performance. Finally, cross-cohort validation and domain adaptation experiments would help determine whether observed effects are stable under distributional shift.

Overall, we hope this benchmark contributes to ongoing efforts in the field toward more structured, reproducible, and variance-aware evaluation of diffusion MRI prediction models. By providing an open and controlled framework, this work aims to complement prior studies and support future investigations that build on shared standards.

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